

BLOOD DONOR AVAILABILITY DASHBOARD

An interactive R shiny web application for public health

SUBMITTED BY

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FOR

Digital Health

Master of science in Global Public Health

DELIVERABLES:

Git hub repository: <https://github.com/janhvi1092/Blood-donor-availability-dashboard->

Shiny application: https://janhvi.shinyapps.io/Blood_donations/

INTRODUCTION:

The report describes the creation and rationale behind the Blood Donor Availability Dashboard; an interactive website created with R and Shiny. Blood donation is an essential part of healthcare systems all around the world. Clinics and blood banks rely on continuous availability of qualified donors who can help meet the needs of those patients who require surgical operations, urgent medical care, or deal with various diseases. However, one of the biggest obstacles in the process of blood donation is the absence of timely information regarding donor demographics, eligibility, and availability of the donor in each centre.

To solve this problem, the authors have developed this dashboard. It allows converting the data about blood donation stored in a CSV file into interactive graphs. With the use of RStudio and Shiny environment, healthcare practitioners, blood bank managers, and officials can analyze the donor information instantly, without having any skills in data analysis. The sample dataset used for the creation of this dashboard includes the data of 10,000 donors at several blood banks. It contains such data as donors' age, gender, blood type, donation status, and eligibility.

KEY USER:

Role:

A Blood Bank Coordinator/Hospital Administrator who will monitor donor information and requires immediate insight into donor eligibility per day and per centre.

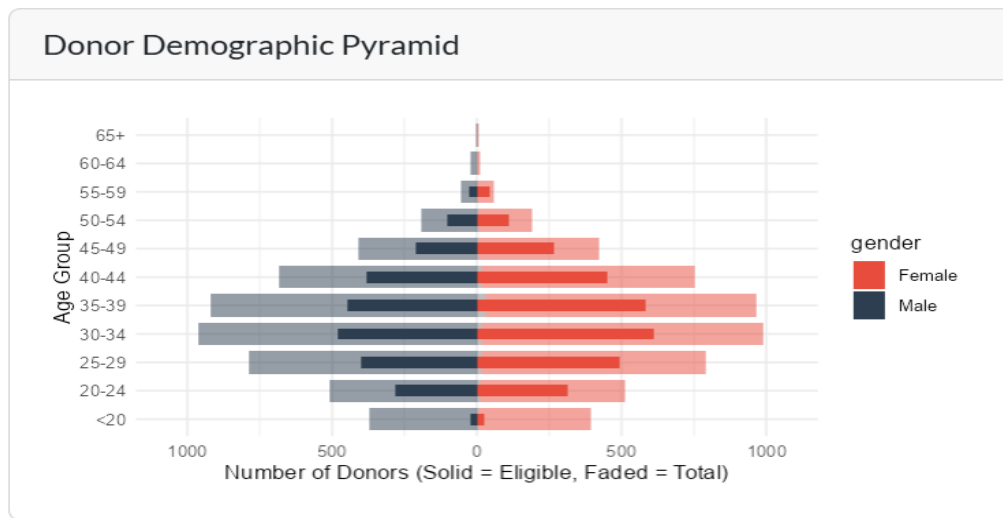
Needs:

- Simple ways of monitoring the number of donors who are eligible per centre per day
- Visual representation of the age and gender composition of the donors to enable them to plan their marketing activities
- A straightforward user interface which does not need programming skills
- The capability of filtering the data according to centres and getting real-time data as output

Aspired Actions:

To analyse the trends of donor demography, plan their marketing activities on the basis of the lower rates of donor eligibility of some age groups and allocate their resources effectively.

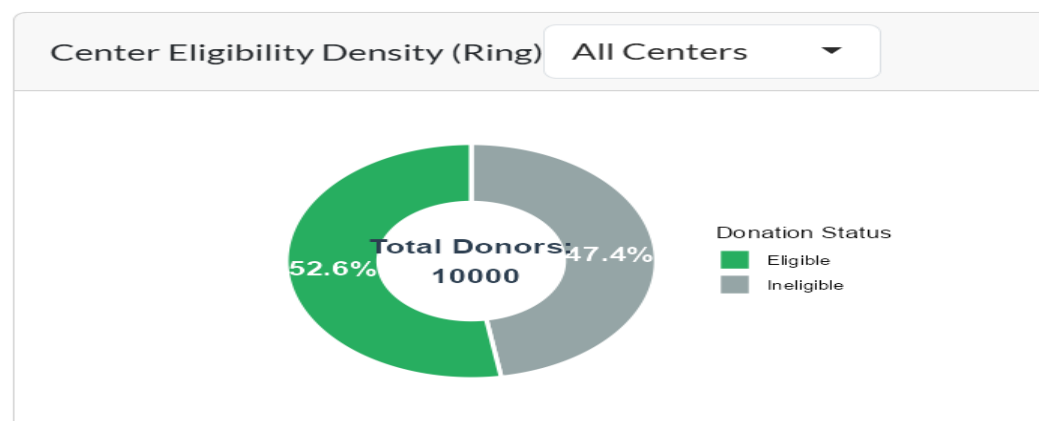
❖ Donor demographic pyramid



Solid bars = Eligible donors | Faded bars = Total donors | Left = Male | Right = Female

This pyramid chart presents the number of donors in terms of their ages and genders. The horizontal bars correspond to different age groups, where the men who donated blood are depicted on the left, and women donors on the right. Darker bars present the number of donors, who are now eligible to donate, whereas lighter bars present the number of total donors (eligible and not eligible) from the same age group. It is clear from the chart that 30-39 age groups are the most active age groups for male and female donors. The number of donors decreases noticeably in the 55+ age groups, which can be explained by the fact that people of older age have higher chances to develop diseases that make them ineligible. Women are more active than men in the 35-44 age group.

❖ CENTRE ELIGIBILITY DENSITY (RING CHART):



Green = Eligible donors (52.6%) | Grey = Ineligible donors (47.4%) | Total = 10,000

Here is the donut ring chart that provides a brief summary of the ratio of eligible donors to ineligible donors at all blood centers. The green sector indicates eligible donors (52.6%) while the grey sector indicates ineligible donors (47.4%). The number of donors (10,000) is displayed in the middle of the ring. One of the insights from the above chart is that almost 50% of the registered donors are not eligible for donating blood. It shows that it is very crucial to keep in mind the point that blood banks cannot solely depend upon the number of registrations while making plans about the supply of the same. The user of the app can easily filter this graph by center through the dropdown menu option.

CONCLUSION:

Through its two primary visual representations: Donor Demographic Pyramid and Centre Eligibility Density Ring Chart, the Blood Donor Availability Dashboard successfully transforms complicated blood donation data into easy-to-understand interactive visuals. The Blood Donor Availability Dashboard offers not only detailed demographics but also a summary level view of the present blood donor availability situation through the two key visuals described above.

The dashboard is made on R and Shiny using ggplot2, plotly, and dplyr packages. It is completely interactive and can be used even by people who do not have technical background. The reactive centre filter ensures that all the graphs get updated immediately after choosing a certain centre. Thus, the dashboard would help hospital administrators decrease time required to analyse donor availability, blood bank coordinators plan their outreach programs, and public health researchers identify trends in demographic aspects of donation. As part of the learning project, the Blood Donor Availability Dashboard serves as an excellent example of meaningful application of data science techniques to solve public health problems.